

Article

Bending–Torsional Coupling Vibration of Hydro-Turbine Generator Unit Considering Gyroscopic Effect Under Multiple Excitations

Zekai Bai ^{1,†}, Jianling Li ^{1,†}, Yunzhe Ma ¹ , Xianan Sun ¹, Hansong Si ¹, Pengchong Zhao ¹, Xianghua Li ¹, Sumin Guan ¹, Bing Peng ¹, Ning Xu ¹, Ziwen Zhao ¹ , Chenchen Song ^{2,*} , Yuhang Yang ¹ and Diyi Chen ^{1,*}

¹ Hubei Technology Innovation Center for Smart Hydropower, Wuhan 430019, China; bzkl8035122306@163.com (Z.B.); jianlingli@nwafu.edu.cn (J.L.); myunzhexy@163.com (Y.M.); sunxn@nwsuaf.edu.cn (X.S.); si_hansong@ctg.com.cn (H.S.); zpc0976@163.com (P.Z.); li_xianghua@ctg.com.cn (X.L.); guan_sumin@ctg.com.cn (S.G.); peng_bing@ctg.com.cn (B.P.); xu_ning3@ctg.com.cn (N.X.); zhaoziwen@nwsuaf.edu.cn (Z.Z.); 2020012056@nwafu.edu.cn (Y.Y.)

² Higher Information Industry Technology Research Institute, Beijing Information Science and Technology University, Beijing 100192, China

* Correspondence: songchenchen@bistu.edu.cn (C.S.); diyichen@nwsuaf.edu.cn (D.C.)

† These authors contributed equally to this work.

Abstract: In this study, a bending–torsional coupling vibration model of the hydro-turbine generator unit (HTGU) incorporating the gyroscopic effect is developed. Using numerical simulation grounded in the actual installation and operational parameters of a hydropower station in China, the vibration characteristics of the HTGU are analyzed under conditions with and without hydraulic excitation, as well as with and without consideration of the gyroscopic effect. Through numerical simulation, it was found that the difference in the x -direction vibration amplitude of the generator rotor between model 1 and model 2 during the start-up phase was less than 5%, indicating that the gyroscopic effect had no influence on the bending–torsional vibration of the HTGU during the start-up phase. However, after a certain period of time, the gyroscopic effect on the bending–torsional vibration of the unit gradually becomes apparent. The vibration amplitude in the x -direction of model 1 is 28% higher than that of model 2, and the amplitude difference in the y -direction reaches 19%. Furthermore, it was found that due to the effects of hydraulic excitation and gyroscopic force, the vibration characteristics of the generator rotor and the turbine runner in the x - and y -directions were different. Under hydraulic excitation, the amplitude of the y -direction vibration of the turbine runner in model 1 was 31% lower than that in model 2. The findings of this study provide a theoretical basis for the modeling and installation of hydro-turbine generator units (HTGUs).

Keywords: the hydro-turbine generator unit (HTGU); the gyroscopic effect; hydraulic excitation; bending–torsional coupling vibration



Academic Editor: Helena M. Ramos

Received: 30 April 2025

Revised: 4 June 2025

Accepted: 6 June 2025

Published: 12 June 2025

Citation: Bai, Z.; Li, J.; Ma, Y.; Sun, X.; Si, H.; Zhao, P.; Li, X.; Guan, S.; Peng, B.; Xu, N.; et al. Bending–Torsional Coupling Vibration of Hydro-Turbine Generator Unit Considering Gyroscopic Effect Under Multiple Excitations. *Water* **2025**, *17*, 1764. <https://doi.org/10.3390/w17121764>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

The hydro-turbine generator unit (HTGU), as the core component of a hydropower plant, serves to convert the mechanical energy of water into electrical energy. Its operational stability is directly related to the economy of a hydropower plant. The HTGU is mainly composed of a rotor, stator, runner, ma (in shaft, upper and lower racks, top cover, three-guide bearing, and thrust bearing. Compared with the general rotating machinery, its structure is complex, so the causes and mechanisms of vibration of the unit are relatively complex [1–3].

The vibration of the HTGU generally includes bending vibration and torsional vibration. Most of the previous studies believe that there is no necessary connection between the bending vibration and torsional vibration [4–6]. But study after study has shown that the bending vibration and torsional vibration are inherent vibration characteristics of the rotor, and there is a possibility of coupling between them [7,8]. The isolated research on bending vibration and torsional vibration is not comprehensive. Starting from engineering practice, studying the bending–torsional coupling vibration of the hydro-turbine generator unit (HTGU) is of substantial significance for preventing major vibration accidents in hydropower plants and ensuring the economic operation of hydropower equipment. At the same time, more auxiliary information conducive to fault diagnosis can be explored from the perspective of vibration coupling [8,9].

At present, in view of the vibration problems existing in the main shaft system of the HTGU, numerous scholars and experts have carried out specific research on the causes of vibration, the vibration reduction measures, and the model optimization of the unit, and they have obtained remarkable results [10–12]. For example, Wu et al. introduced a 1D convolutional neural network (CNN)-based methodology to automatically derive effective features for rub-impact fault diagnosis from raw vibration signals of a rotor system, and they accomplished the detection of rub-impact faults via a straightforward 1D CNN structure [13]. Xu et al. investigated the effect of nonlinear unbalanced electromagnetic pull (UMP) on the radial vibration of a large hydro-turbine generator and computed the dynamic responses of the finite element model under various analytical conditions for UMP [4]. Yang et al. proposed a methodology for parameterized time–frequency analysis (TFA) of the nonstationary vibration signal of variable-speed rotating machinery, and the effectiveness of the proposed method in analyzing mono-component and multi-component signals was demonstrated [14]. These studies are aimed at the unbalanced electromagnetic pull and nonlinear oil film force during the operation of the HTGU, and the hydraulic excitation of the HTGS, as special rotating machinery, has not received much attention. In addition, Xu et al. developed a bending vibration model incorporating the penstock, power grid, and guide vane device and analyzed the influences of coupled hydro-mechano-electrical factors [15]. Song et al. investigated the torsional vibration characteristics of the hydro-turbine generator unit, with consideration of the coupling between electromagnetic and hydraulic vibration sources, and analyzed the effects of the excitation current and internal active power angle on torsional vibration [11]. Xu et al. developed a finite element rotor dynamic model for radial vibration considering operational conditions and proposed a novel algorithm capable of addressing both transient and steady-state behaviors [16]. Zhou et al. conducted a study on bidirectional fluid–solid interaction (BFSI) and a detailed analysis of the transient forces of the impeller and the vibration characteristics of the rotor system when BFSI was considered under different flow velocities, wear ring gaps, and axial movements [17]. However, most of these studies have concentrated on the vibration of a single rotor and the bending vibration of the hydro-turbine generator unit, whereas investigations into the bending–torsional coupling vibration of the HTGU remain relatively scarce. Recently, a comprehensive analysis of the bending–torsional coupling vibration characteristics of the hydro-turbine generator unit has been performed. For instance, Shi et al. analyzed the vibrational stability and shafting vibration sensitivity to specific parameters in the unbalanced rotor bending–torsional coupling vibration of the hydro-turbine generator unit and discussed the effect of generator mass eccentricity on the generator rotor and turbine runner [18]. Hua et al. developed a nonlinear dynamic model for rotor systems that couples bending and torsional vibrations with consideration of nonlinear friction, and investigated the coupled vibration characteristics of the rotor system under nonlinear friction conditions [19]. Sun et al. proposed a method for studying the

overall nonlinear dynamics of the hydraulic generator shaft–base coupling system, which has enriched the knowledge of its dynamic behavior [20]. These research outcomes further advance the investigation into the bending–torsional coupling vibration of the HTGU shaft system. Nevertheless, the aforementioned studies did not account for the impacts of the gyroscopic effect and hydraulic excitation on HTGU vibration. In practical engineering operations, as hydro-turbine generator units rapidly transition toward a large capacity, high head, and high speed, the influences of the gyroscopic effect and hydraulic excitation on HTGUs cannot be overlooked [21,22]. Therefore, to further optimize the vibration model of the HTGU, it is essential to clarify the multiple driving forces behind the unit’s vibration at a detailed level, thereby establishing a nonlinear dynamic model of the HTGU that better aligns with engineering practices. This establishes a robust foundation for analyzing vibration issues in the HTGU and offers a theoretical framework for fault diagnosis and prevention of the unit.

Based on the above discussion, this study focus on the bending–torsional coupling vibration characteristics of the gyroscopic effect and the hydraulic excitation on the HTGU by establishing the bending–torsional coupling vibration nonlinear mathematical model. Compared with previous studies, this study has the following three main innovations. First, a bending–torsional coupling vibration nonlinear mathematical model of the hydro-turbine generator unit incorporating the gyroscopic effect and external excitation (including hydraulic excitation) is proposed innovatively. This model is more suitable for engineering practice and development demand. Second, a comparison model is established in this paper, and the effect of hydraulic excitation is comprehensively considered on the basis of investigating the impact of the gyroscopic effect on the vibration of the HTGU. The vibrational characteristics of the HTGU in the presence of the gyroscopic effect and hydraulic excitation are analyzed. Third, from the engineering point of view, combined with the development trend of domestic hydropower, the vibration issues of the HTGU under large-capacity, high-head, and high-speed conditions are investigated, which provides a theoretical basis for further ensuring the safe and stable operation of the HTGU.

The organizational structure of this paper is as follows: Section 2 proposes a nonlinear mathematical model for bending–torsional coupling vibration that comprehensively considers multiple influencing factors to represent most practical scenarios. Section 3 is based on the actual installation and operational parameters of a Chinese hydropower station to analyze the vibration characteristics of the generator rotor and turbine runner in the x - and y -directions under the combined influence of gyroscopic forces and hydraulic excitation, as well as the influence of including or excluding the gyroscopic effect under different operating conditions. Section 4 presents conclusions that further demonstrate the significant impact of gyroscopic effects and hydraulic excitation on the HTGU during hydro-power plant operation, which cannot be overlooked.

2. Mathematical Modeling of Bending–Torsional Coupled Vibration for the HTGU

2.1. Gyroscopic Moment

Rotating objects and non-rotating objects have different dynamic behaviors under the influence of the gyroscopic moment (GE). The GE has a significant influence on the rigid modal frequency of the rotor system when the rotational inertia J_d of the x - and y -axes is less than the z -axis rotational inertia J_p . Therefore, in the equation of motion, the influence of the GE must be taken into account.

Assuming the rotor mass is m , under the action of impact force F and impact moment M , the instantaneous change in momentum P can be expressed as [23]

$$\Delta P = \int F dt \quad (1)$$

The impact moment M can be written as

$$M = d \times F \quad (2)$$

The initial momentum moment of rotor can be written as

$$L_0 = J_p \Omega \quad (3)$$

where Ω denotes the angular velocity of the rotor rotating about the z -axis, and J_p represents the rotor's polar moment of inertia. The relationships among the variables are shown in the Figure 1 below.

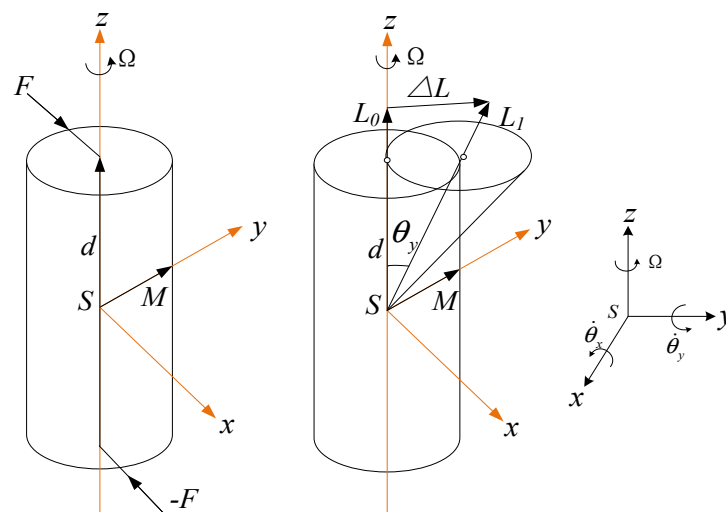


Figure 1. Schematic of the gyroscopic effect.

The rotor is in a constant axis rotation in the initial state, and the main shaft, rotation shaft, and rotor's moment axis are coincident. Under the influence of impact moment M , the rotor spins along the y -axis with a certain angle θ_y , and the initial momentum moment changes from L_0 to L_1 . Thus, the variation can be written as [24]

$$\Delta L = \int M dt \quad (4)$$

Based on the momentum conservation principle, the rotor is subject to a moment of inertia with an equal magnitude and opposite direction, i.e., the gyroscopic moment. It can be expressed as

$$M_x = -\Delta L = -J_p \Omega \dot{\theta}_y \quad (5)$$

where the negative sign indicates the direction of the gyroscopic moment along the negative direction of the x -axis. $\dot{\theta}_y$ is the angular velocity of rotation in the positive direction of the y -axis.

In the same way, the gyroscopic moment of the rotor in the x -axis direction can be obtained as

$$M_x = J_p \Omega \dot{\theta}_x \quad (6)$$

where $\dot{\theta}_x$ is the angular velocity in the direction of the x -axis.

2.2. Euler Angles

The rotor is considered as a disc, and a fixed coordinate system $O-xyz$ is established with O as the origin. In addition, the coordinate system $O-x_2y_2z_2$, which is fixed on the disc and moves with it, is established. The disc is located at the position $O-x_0y_0z_0$ and coincides with $O-xyz$ at the beginning. When rotating, the rotor rotates around the Oy -axis at an angle of θ_y to $O-x_1y_1z_1$, and then rotates around the Ox_1 -axis at an angle of θ_x to $O-x_1y_1z_2$; finally, it turns around the Oz_2 -axis to $O-x_2y_2z_2$ [25]. The illustration of Euler angles is shown in Figure 2.

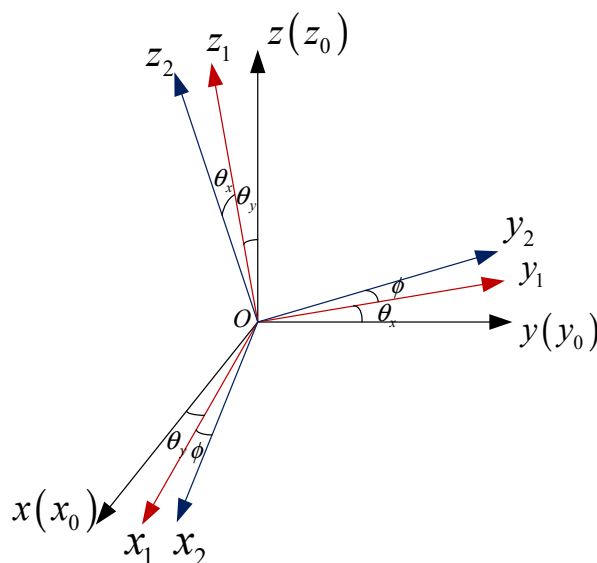


Figure 2. The illustration of Euler angles.

Assuming that the rotational speed of the rotating coordinate system around the translational coordinate system are $\dot{\theta}_x$, $\dot{\theta}_y$, and $\dot{\phi}$, respectively, the absolute rotational angular velocity of the rotating coordinate system can be expressed as

$$\omega = \dot{\theta}_x + \dot{\theta}_y + \dot{\phi} \quad (7)$$

The disc's rotation is decomposed into rotation with the coordinate system $O-x_1y_1z_2$ and rotation relative to this coordinate system. Subsequently, the rotational angular velocity of the coordinate system $O-x_1y_1z_2$ may be expressed as

$$\omega_1 = \dot{\theta}_x + \dot{\theta}_y \quad (8)$$

The rotation of the disc relative to this moving coordinate system is the rotation around the central axis, and $\dot{\phi} = \Omega$ is termed the rotational angular velocity. The projection of angular velocity ω on each axis of $O-x_1y_1z_2$ can be described as

$$\begin{cases} \omega_{x1} = \dot{\theta}_x \\ \omega_{y1} = \dot{\theta}_y \cos(\theta_x) \\ \omega_{z2} = \dot{\phi} - \dot{\theta}_y \sin(\theta_x) \end{cases} \quad (9)$$

The direction cosine matrix from the coordinate system $Ox_1y_1z_2$ to the coordinate system $Ox_2y_2z_2$ can be written as

$$A = \begin{bmatrix} \cos(\phi) & \sin(\phi) & 0 \\ -\sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (10)$$

Therefore, the projection of the angular velocity ω on each axis of $Ox_2y_2z_2$ can be expressed as

$$\begin{bmatrix} \omega_{x2} \\ \omega_{y2} \\ \omega_{z2} \end{bmatrix} = \begin{bmatrix} \cos(\phi) & \sin(\phi) & 0 \\ -\sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_x \\ \dot{\theta}_y \cos(\theta_x) \\ \dot{\phi} - \dot{\theta}_y \sin(\theta_x) \end{bmatrix} = \begin{bmatrix} \dot{\theta}_x \cos(\phi) + \dot{\theta}_y \cos(\theta_x) \sin(\phi) \\ -\dot{\theta}_x \sin(\phi) + \dot{\theta}_y \cos(\theta_x) \sin(\phi) \\ \dot{\phi} - \dot{\theta}_y \sin(\theta_x) \end{bmatrix} \quad (11)$$

In light of above discussion, the rotational kinetic energy of the rotor system can be expressed as

$$\begin{aligned} T_R &= \frac{1}{2} \left[J_{p1} (\omega_{x2}^2 + \omega_{y2}^2) + J_d \omega_{z2}^2 \right] \\ &= \frac{1}{2} \left[J_d \left(\dot{\theta}_x^2 + \dot{\theta}_y^2 \cos^2 \theta_x \right) + J_{p1} \dot{\phi}_1^2 + J_{p1} \dot{\theta}_y^2 \sin^2 \theta_x - 2J_{p1} \dot{\phi}_1 \dot{\theta}_y \sin \theta_x \right] \end{aligned} \quad (12)$$

2.3. The HTGU Bending–Torsional Coupled Vibration Model Considering Gyroscopic Effects

Figure 3 shows the structural diagram of the unit shafting system and the vibration calculation sketch. In this context, $O_1 (x_1, y_1)$ and $O_2 (x_2, y_2)$ denote the geometric centers of the generator rotor and hydraulic turbine runner, respectively. The masses of the generator rotor and hydraulic turbine runner are represented by m_1 and m_2 , correspondingly. The angles of rotation for the generator rotor and hydraulic turbine runner are denoted ϕ_1 and ϕ_2 , respectively. Meanwhile, e_1 and e_2 signify the mass eccentricities of the rotor and runner, respectively.

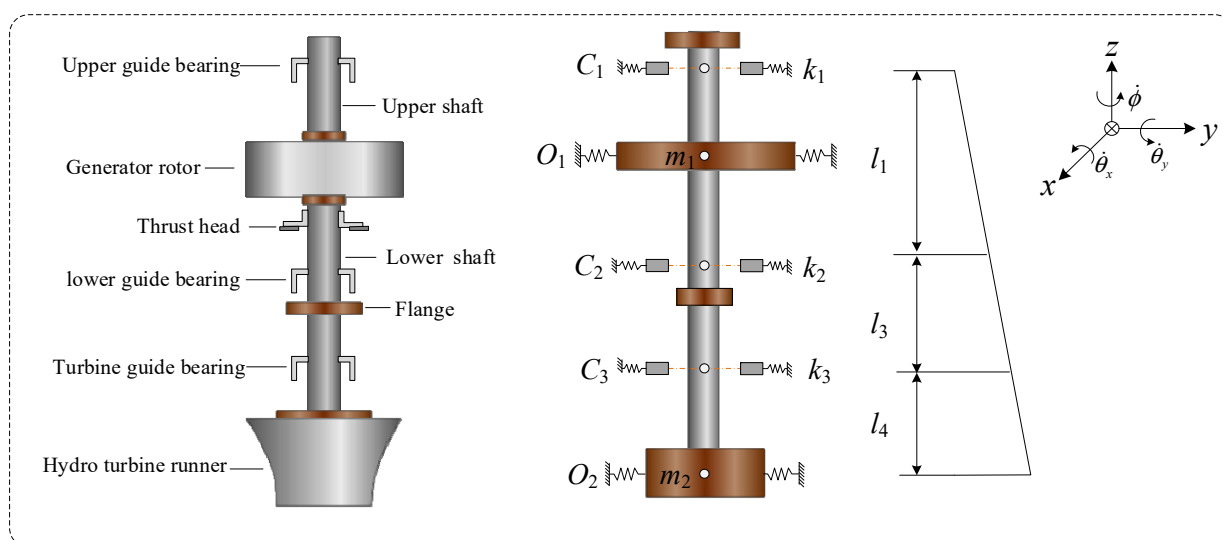


Figure 3. Schematic diagram of the shafting structure and the vibration calculation model.

$O_1 (x_1, y_1)$ represents the geometric centers of the generator rotor and $O_2 (x_2, y_2)$ indicates the hydraulic turbine runner, respectively; and the masses of the generator rotor and hydraulic turbine runner are denoted m_1 and m_2 , respectively. The rotational angles of the generator rotor and hydraulic turbine runner are represented by ϕ_1 and ϕ_2 ,

correspondingly. Additionally, the mass eccentricities of the rotor and runner are given by e_1 and e_2 , respectively.

The shafting system can be simplified as a two-disc bending–torsional coupling vibration system composed of a rotor and a runner. In view of the above discussion, the total kinetic energy of the HTGU can be expressed as

$$\begin{aligned}
 T = & \frac{1}{2}m_1 \left(\dot{x}_1^2 + \dot{y}_1^2 + e_1^2 \dot{\phi}_1^2 + 2e_1 \dot{\phi}_1 \dot{y}_1 \cos \phi_1 - 2e_1 \dot{\phi}_1 \dot{x}_1 \sin \phi_1 \right) + \frac{1}{2}m_1 e_1^2 \dot{\phi}_1^2 \\
 & + \frac{1}{2} \left[J_d \left(\dot{\theta}_x^2 + \dot{\theta}_y^2 \cos^2 \theta_x \right) + J_{p1} \dot{\phi}_1^2 + J_{p1} \dot{\theta}_y^2 \sin^2 \theta_x - 2J_{p1} \dot{\phi}_1 \dot{\theta}_y \sin \theta_x \right] \\
 & + \frac{1}{2}m_2 \left(\dot{x}_2^2 + \dot{y}_2^2 + e_2^2 \dot{\phi}_2^2 + 2e_2 \dot{\phi}_2 \dot{y}_2 \cos \phi_2 - 2e_2 \dot{\phi}_2 \dot{x}_2 \sin \phi_2 \right) + \frac{1}{2}(J_{p2} + m_2 e_2^2) \dot{\phi}_2^2
 \end{aligned} \quad (13)$$

The overall potential energy of the HTGU can be written as [26]

$$\begin{aligned}
 U = & K_{11}(x_1^2 + y_1^2) + K_{22}(x_2^2 + y_2^2) + K_{12}\sqrt{x_1^2 + y_1^2}\sqrt{x_2^2 + y_2^2} \\
 & + \mu(\theta_y x_1 + \theta_x y_1) + \frac{1}{2}\chi(\theta_x^2 + \theta_y^2) + \frac{1}{2}k_t(\phi_1 - \phi_2)^2
 \end{aligned} \quad (14)$$

Among them,

$$\begin{aligned}
 K_{11} = & \frac{2k_1(l_1+l_3+l_4)^2+2k_2(l_3+l_4)^2+2k_3l_4^2}{(l_1+2l_3+2l_4)^2} \quad K_{22} = \frac{k_1l_1^2+k_2l_1^2+k_3(l_1+2l_3)^2}{2(l_1+2l_3+2l_4)^2} \\
 K_{12} = & \frac{2l_1k_1(l_1+l_3+l_4)+2l_1k_2(l_3+l_4)+2l_4k_3(l_1+2l_3)}{(l_1+2l_3+l_4)^2}
 \end{aligned}$$

In this expression, the support stiffness values of the upper guide bearing, lower guide bearing, and water guide bearing are denoted k_1 , k_2 , and k_3 , respectively. The spring constant of the main shaft under the action of force F is represented by μ , while the spring constant of the main shaft under torque M is given by χ . Additionally, k_t signifies the torsional stiffness of the shafting.

By substituting Equations (13) and (14), the Lagrangian of the system is derived as

$$\begin{aligned}
 L = & T - U \\
 = & \frac{1}{2}m_1 \left(\dot{x}_1^2 + \dot{y}_1^2 + e_1^2 \dot{\phi}_1^2 + 2e_1 \dot{\phi}_1 \dot{y}_1 \cos \phi_1 - 2e_1 \dot{\phi}_1 \dot{x}_1 \sin \phi_1 \right) + \frac{1}{2}m_1 e_1^2 \dot{\phi}_1^2 \\
 & + \frac{1}{2} \left[J_d \left(\dot{\theta}_x^2 + \dot{\theta}_y^2 \cos^2 \theta_x \right) + J_{p1} \dot{\phi}_1^2 + J_{p1} \dot{\theta}_y^2 \sin^2 \theta_x - 2J_{p1} \dot{\phi}_1 \dot{\theta}_y \sin \theta_x \right] \\
 & + \frac{1}{2}m_2 \left(\dot{x}_2^2 + \dot{y}_2^2 + e_2^2 \dot{\phi}_2^2 + 2e_2 \dot{\phi}_2 \dot{y}_2 \cos \phi_2 - 2e_2 \dot{\phi}_2 \dot{x}_2 \sin \phi_2 \right) + \frac{1}{2}(J_{p2} + m_2 e_2^2) \dot{\phi}_2^2 \\
 & - \left[K_{11}(x_1^2 + y_1^2) + K_{22}(x_2^2 + y_2^2) + K_{12}\sqrt{x_1^2 + y_1^2}\sqrt{x_2^2 + y_2^2} + \mu(\theta_y x_1 + \theta_x y_1) \right. \\
 & \quad \left. + \frac{1}{2}\chi(\theta_x^2 + \theta_y^2) + \frac{1}{2}k_t(\phi_1 - \phi_2)^2 \right]
 \end{aligned} \quad (15)$$

In this paper, the generalized coordinates are chosen as $q_i = \{x_1, y_1, \phi_1, \theta_x, \theta_y, x_2, y_2, \phi_2\}$, and all external forces exerted on the hydro-turbine generator unit are noted as $F = \sum F_{ij}$ ($i = x, y, \phi, \theta_x, \theta_y, j = 1, 2$). The Lagrange equation for the system can be expressed as

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \sum F_{ij} \quad (16)$$

The external forces acting on the HTGU can be written as

$$\left\{ \begin{array}{l} \sum F_{x1} = F_{x-ump} - c_1 \dot{x}_1 \\ \sum F_{y1} = F_{y-ump} - c_1 \dot{y}_1 \\ \sum F_{\phi 1} = M_e - c_t \dot{\phi}_1 \\ \sum F_{\theta x} = -c_{r\theta} \dot{\theta}_x \\ \sum F_{\theta y} = -c_{r\theta} \dot{\theta}_y \\ \sum F_{x2} = F_{x-oil} + F_{\tau yx} - c_2 \dot{x}_2 \\ \sum F_{y2} = F_{y-oil} + F_{\tau yy} - c_2 \dot{y}_2 \\ \sum F_{\phi 2} = M - c_t \dot{\phi}_2 \end{array} \right. \quad (17)$$

where the damping forces for the generator rotor and turbine runner in the x - and y -directions are represented by $c_1 \dot{x}_1$, $c_1 \dot{y}_1$ and $c_2 \dot{x}_2$, $c_2 \dot{y}_2$, respectively; axial torsional damping of the generator rotor and turbine runner is denoted by $c_t \dot{\phi}_1$ and $c_t \dot{\phi}_2$, respectively; damping torques of the generator rotor in the x - and y -directions are expressed as $c_{r\theta} \dot{\theta}_x$ and $c_{r\theta} \dot{\theta}_y$, respectively; the x - and y -directions components of the unbalanced magnetic pulling force acting on the generator rotor are given by F_{x-ump} and F_{y-ump} , respectively; oil film forces acting on the turbine runner in the x - and y -directions are represented by F_{x-oil} and F_{y-oil} , respectively; M_e signifies the electromagnetic torque applied to the generator rotor; and M corresponds to the hydraulic excitation couple moment acting on the turbine runner.

(1) Unbalanced magnetic pull

The vibration characteristics of the hydro-turbine generator unit are significantly influenced by the unbalanced magnetic pull. An uneven air gap between the stator and rotor, resulting from factors such as the imbalanced mass of the generator rotor, initial shaft deflection, or hydraulic imbalance, subjects the rotor to a lateral unbalanced magnetic pull. This, in turn, exacerbates the unit's vibration. The unbalanced magnetic pull can be expressed as [15]

$$\left\{ \begin{array}{l} F_{x-ump} = \frac{RL\pi k_j^2 I_j^2}{4\mu_0} (2\Lambda_0\Lambda_1 + \Lambda_1\Lambda_2 + \Lambda_2\Lambda_3) \cos \gamma \\ F_{y-ump} = \frac{RL\pi k_j^2 I_j^2}{4\mu_0} (2\Lambda_0\Lambda_1 + \Lambda_1\Lambda_2 + \Lambda_2\Lambda_3) \sin \gamma \end{array} \right. \quad (18)$$

In this context, R denotes the radius of generator rotor; L represents the length of the generator rotor; δ_0 signifies the gap between stator and rotor; and γ is the rotation angle, defined by $\cos \gamma = \frac{x_1}{e}$ and $\sin \gamma = \frac{y_1}{e}$. $e = \sqrt{x_1^2 + y_1^2}$ corresponds to the radial amplitude of the generator rotor; $\varepsilon = \frac{e}{\delta_0}$ represents the relative eccentricity; μ_0 is the air permeability; I_j denotes the excitation current of the generator rotor; k_j is the fundamental factor of magnetomotive force; and Λ_0 , Λ_1 , Λ_2 , and Λ_3 are four intermediate variables, which can be expressed as

$$\left\{ \begin{array}{l} \Lambda_0 = \frac{\mu_0}{\delta_0} \frac{1}{\sqrt{1-\varepsilon^2}} \\ \Lambda_1 = \frac{2\mu_0}{\delta_0} \frac{1}{\sqrt{1-\varepsilon^2}} \left(\frac{1-\sqrt{1-\varepsilon^2}}{\varepsilon} \right) \\ \Lambda_2 = \frac{2\mu_0}{\delta_0} \frac{1}{\sqrt{1-\varepsilon^2}} \left(\frac{1-\sqrt{1-\varepsilon^2}}{\varepsilon} \right) \\ \Lambda_3 = \frac{2\mu_0}{\delta_0} \frac{1}{\sqrt{1-\varepsilon^2}} \left(\frac{1-\sqrt{1-\varepsilon^2}}{\varepsilon} \right) \end{array} \right. \quad (19)$$

(2) Oil film force

The formula for the oil film force on the shaft diameter may be expressed as [27]

$$\left\{ \begin{array}{l} F_{x-oil} = F_{x0} + k_{xx}x_2 + k_{xy}y_2 + d_{xx}\dot{x}_2 + d_{xy}\dot{y}_2 \\ F_{y-oil} = F_{y0} + k_{yx}x_2 + k_{yy}y_2 + d_{yx}\dot{x}_2 + d_{yy}\dot{y}_2 \end{array} \right. \quad (20)$$

where F_{x0} and F_{y0} denote the static oil film forces in the x - and y -directions at the axis diameter, respectively; and k_{xx} , k_{xy} , k_{yx} , k_{yy} , d_{xx} , d_{xy} , d_{yx} , and d_{yy} are intermediate variables whose expressions can be written as

$$\left\{ \begin{array}{l} k_{xx} = \left(\frac{B}{d}\right)^2 \frac{4\epsilon' [2\pi^2 + (16 - \pi^2)\epsilon'^2]}{(1 - \epsilon'^2)^2 [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \\ k_{xy} = \left(\frac{B}{d}\right)^2 \frac{\pi [-\pi^2 + 2\pi^2\epsilon'^2 + (16 - \pi^2)\epsilon'^4]}{(1 - \epsilon'^2)^{\frac{5}{2}} [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \\ k_{yx} = \left(\frac{B}{d}\right)^2 \frac{\pi [\pi^2 + (\pi^2 + 32)\epsilon'^2 + 2(16 - \pi^2)\epsilon'^4]}{(1 - \epsilon'^2)^{\frac{5}{2}} [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \\ k_{yy} = \left(\frac{B}{d}\right)^2 \frac{4\epsilon' [\pi^2 + (\pi^2 + 32)\epsilon'^2 + 2(16 - \pi^2)\epsilon'^4]}{(1 - \epsilon'^2)^3 [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \\ d_{xx} = \left(\frac{B}{d}\right)^2 \frac{2\pi [\pi^2 + 2\pi^2\epsilon'^2 - 16\epsilon'^2]}{(1 - \epsilon'^2)^{\frac{3}{2}} [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \\ d_{xy} = \left(\frac{B}{d}\right)^2 \frac{8\epsilon' [\pi^2 + 2\pi^2\epsilon'^2 - 16\epsilon'^2]}{(1 - \epsilon'^2)^2 [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \\ d_{yx} = d_{xy} \\ d_{yy} = \left(\frac{B}{d}\right)^2 \frac{2\pi [\pi^2 + 2\pi^2\epsilon'^2 - 16\epsilon'^2]}{(1 - \epsilon'^2)^2 [16\epsilon'^2 + \pi^2(1 - \epsilon'^2)]} \end{array} \right. \quad (21)$$

In this paper, B/d represents the bearing width-to-diameter ratio; and $\epsilon' = \frac{\sqrt{x_2^2 + y_2^2}}{c}$ denotes the eccentricity ratio, where c is the radial bearing clearance.

(3) Hydraulic excitation

When a hydro-turbine generator set operates normally, the turbine runner is primarily propelled by the fluid's rotational thrust against the blades. The radial gap excitation force of flow and the unbalanced moment of coupling by the flow are introduced into the model. The hydraulic excitation of the turbine runner can be written as

$$\left\{ \begin{array}{l} F_{\tau yx} = -\xi_0 Q \frac{\rho}{A} (v_2 \cos(\beta) - v_1 \cos(\alpha)) (x_2^2 + y_2^2) \cos(\Omega t) \\ F_{\tau yy} = -\xi_0 Q \frac{\rho}{A} (v_2 \cos(\beta) - v_1 \cos(\alpha)) (x_2^2 + y_2^2) \sin(\Omega t) \end{array} \right. \quad (22)$$

where Ω is the excitation frequency; for the turbine runner, v_1 and v_2 represent the flow velocities at the inlet and outlet, respectively; the symbols α and β denote the inflow angle and outflow angle, respectively; ξ_0 is the disturbance coefficient; A is the area of the turbine runner; and ρ is the density of water.

(4) Electromagnetic torque

According to the reference [11], the energy associated with the air-gap field of the 0020 generator rotor can be formulated as

$$W = \frac{R_g L' \Lambda_0}{2} \int_0^{2\pi} \left(1 + \frac{x_1^2 + y_1^2}{2\sigma^2} + \frac{x_1}{\sigma} \cos \alpha_1 + \frac{y_1}{\sigma} \sin \alpha_1 + \frac{x_1^2 - y_1^2}{2\sigma^2} \cos 2\alpha_1 + \frac{x_1 y_1}{\sigma^2} \sin 2\alpha_1 \right) \times \left[F_{sm} \cos(\omega_1 t - p\alpha) + F_{rm} \cos\left(\omega_1 t - p\alpha + \varphi + \gamma + \frac{\pi}{2}\right) \right]^2 d\alpha \quad (23)$$

In this context, F_{sm} and F_{rm} denote the magnitudes of the fundamental magnetic potential waves in the stator and rotor windings, respectively. The parameter γ corresponds to the internal power angle, while φ represents the rotor torsion angle. R_g signifies the inner radius of the generator stator, and L' denotes the effective length of the generator rotor. Λ_0 indicates the permeance of the uniform air gap in the generator. The term σ is defined as $k_\mu \delta_0$, where k_μ represents the saturation factor and δ_0 denotes the length of the uniform air gap. α_1 is the angle between the x -axis and the air-gap position, whereas β_1 corresponds to the angle of the x -coordinate at the minimum air-gap location. ω_1 represents the synchronous rotational speed, and p stands for the quantity of magnetic pole pairs.

The air-gap magnetic field energy can be expressed as

$$F(\alpha, t) = F_{sm} \cos(\omega t - p\alpha) + F_{rm} \cos\left(\omega t - p\alpha + \varphi + \gamma + \frac{\pi}{2}\right) \quad (24)$$

The partial derivative of the rotor torsional vibration angle is derived using Equation (24), which provides a mathematical framework. When omitting the higher-order term of φ because the amplitude of the nonlinear term of the electromagnetic torque is small, the expression of the electromagnetic torque can be obtained as

$$M_e = -2\pi R_0 \cos \gamma + \left[R_0 R_1 - \frac{R_0}{2} \sin(2\omega t + \gamma) \right] \varphi + \frac{R_0}{2p} \cos(2\omega t + \gamma) \quad (25)$$

where $R_0 = \frac{RLp\Lambda_0 R_m}{2} \left(1 + \frac{e^2}{2\sigma^2}\right)$, $R_1 = 2p\pi \sin \gamma$, and $R_m = F_{sm} F_{rm}$.

(5) Couple moment of hydraulic excitation

The force couple of the flow on the turbine runner includes the constant force couple caused by the constant flow and the abnormal disturbance caused by the unsteady flow or other factors [28]. Assumption:

$$M = M_0 + f_0(t) \quad (26)$$

where M_0 is a constant flow and $f_0(t)$ is considered pulsation with an amplitude of 0.18, i.e.,

$$f_0(t) = 0.18 M_0 f(t) \quad (27)$$

where $f(t)$ is a periodic function with a frequency of ω_m .

The constant force couple acting on the turbine runner as a result of the constant flow can be expressed as

$$M_0 = \frac{r}{g} Q(v_2 r_2 - v_1 r_1) \quad (28)$$

where r denotes the maximum distance from the axis of the runner; Q is the effective flow through the turbine runner; and $v_2 r_2$ and $v_1 r_1$ denote the velocity moments at the inlet and outlet of the runner.

In summary, the couple moment of hydraulic excitation can be expressed as

$$M = M_0 + f_0(t) = \frac{r}{g} Q(v_2 r_2 - v_1 r_1) + 0.18 M_0 f(t) \quad (29)$$

By substituting Equations (15) and (17) into Equation (16), the bending–torsional coupling vibration model of the shafting system with the gyroscopic effect (model 1) can be obtained as [3,29]

$$\text{model 1} \begin{cases} m_1 \ddot{x}_1 + c_{sys} \dot{x}_1 + k_{sys} x_1 = f_{1x} + f_{2x} + f_{3x} + f_{6x} \\ m_1 \ddot{y}_1 + c_{sys} \dot{y}_1 + k_{sys} y_1 = f_{1y} + f_{2y} + f_{3y} + f_{6y} \\ J_{p1} \ddot{\phi}_1 + c_t \dot{\phi}_1 + k_t \phi_1 = f_4 + f_5 \end{cases} \quad (30)$$

Here, f_1 represents the unbalanced magnetic force (Equation (18)), f_2 represents the oil film force (Equation (20)), f_3 represents the hydraulic excitation force (Equation (22)), f_4 represents the electromagnetic torque (Equation (25)), f_5 represents the hydraulic excitation torque (Equation (29)), represents the gyroscopic torque term (Equations (5) and (6)), c_{sys} represents the combined damping term, and k_{sys} represents the combined stiffness term.

Moreover, to assess the impact of the gyroscopic effect on the bend–torsional coupled vibration of the hydro-turbine generator unit (HTGU) shafting system, this study introduces

a comparative model (model 2) for the shafting system's bend–torsional coupling vibration that excludes the gyroscopic effect. This comparative modeling approach facilitates a systematic evaluation of the gyroscopic effect's role in the dynamic behavior of the HTGU shafting system's coupled bending and torsional vibrations. Model 2 can be written as

$$\text{model 2} \begin{cases} m_1 \ddot{x}_1 + c_{sys} \dot{x}_1 + k_{sys} x_1 = f_{1x} + f_{2x} + f_{3x} \\ m_1 \ddot{y}_1 + c_{sys} \dot{y}_1 + k_{sys} y_1 = f_{1y} + f_{2y} + f_{3y} \\ J_{p1} \ddot{\phi}_1 + c_t \dot{\phi}_1 + k_t \phi_1 = f_4 + f_5 \end{cases} \quad (31)$$

3. Numerical Analysis

A detailed analysis of the bending–torsion coupling vibration characteristics of the shafting system during the operation of the unit is carried out in conjunction with the actual installation parameters of a hydropower unit. The motion equation established in this paper is solved using the fourth-order Runge–Kutta numerical integration method. This method has high computational accuracy and good stability, and it is suitable for solving the nonlinear differential equation systems in this study, which involve multiple physical field couplings such as gyroscopic effects and hydrodynamic excitation. The solution process is as follows.

First, the high-order differential equation system is transformed into a first-order differential equation system form, and a state vector including variables such as displacement, velocity, twist angle, and angular velocity is constructed to facilitate iterative numerical calculation. Secondly, the initial conditions and time step are set. The initial value of model 1 is [0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.1; 0.0001] and the initial value of model 2 is [0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.001; 0.0001; 0.1; 0.0001]. Finally, the adaptive step size control strategy was implemented using the ode45 solver of the MATLAB R2024a (version 9.16.0; The MathWorks, Natick, MA, USA). During the smooth response stage of the system, a larger step size was used to improve efficiency. When the vibration changed drastically (such as during the start-up transition process), the step size was automatically reduced to ensure numerical stability. The state variables at each time step were calculated through iterative computations. The actual parameters of the hydropower station, including the rotor mass of the generator, the turbine runner mass, and the stiffness of the bearing support, were used to ensure the engineering practicability of the calculation results. The basic parameters of the shafting system are shown in Table 1.

Table 1. The basic parameters of the shafting system of the HTGU.

Parameters	Values	Units	Parameters	Values	Units
m_1	4.5×10^5	kg	k_1	8.5×10^7	N/m
m_2	2.5×10^5	kg	k_2	6.5×10^7	N/m
c_1	4.5×10^5	N·s/m	k_3	3.5×10^7	N/m
c_2	3.5×10^5	N·s/m	K_t	7.5×10^6	N/m
c_t	2.5×10^5	N·s/m	cr_0	2.5×10^4	N·s/m
e_1	0.5×10^{-3}	m	μ	3.5×10^7	N/m
e_2	0.5×10^{-3}	m	χ	4.5×10^7	N/m
J_d	4.5×10^7	kg·m ²	v_1	12	m/s
J_{p1}	6.5×10^7	kg·m ²	v_2	11	m/s
J_{p2}	1.5×10^6	kg·m ²	R	0.7	m

Table 1. Cont.

Parameters	Values	Units	Parameters	Values	Units
α	11.43	$^{\circ}$	r	1.6	m
β	12.47	$^{\circ}$	r_1	1.2	m
Q	960	m^3/s	r_2	0.8	m
L	0.4	m	δ_0	2×10^{-3}	m
A	8.04	m^2	k_u	1.2	p.u
ρ	1×10^3	kg/m^3			

3.1. The Impact of the Gyroscopic Effect on the Bending–Torsional Coupled Vibration of the HTGU with Hydraulic Excitation

The time histories and the frequency spectra of the generator rotor and turbine runner along the x - and y -directions are shown in Figures 4 and 5, where model 1 indicates that the gyroscopic effect is considered and model 2 indicates that the gyroscopic effect is not considered.

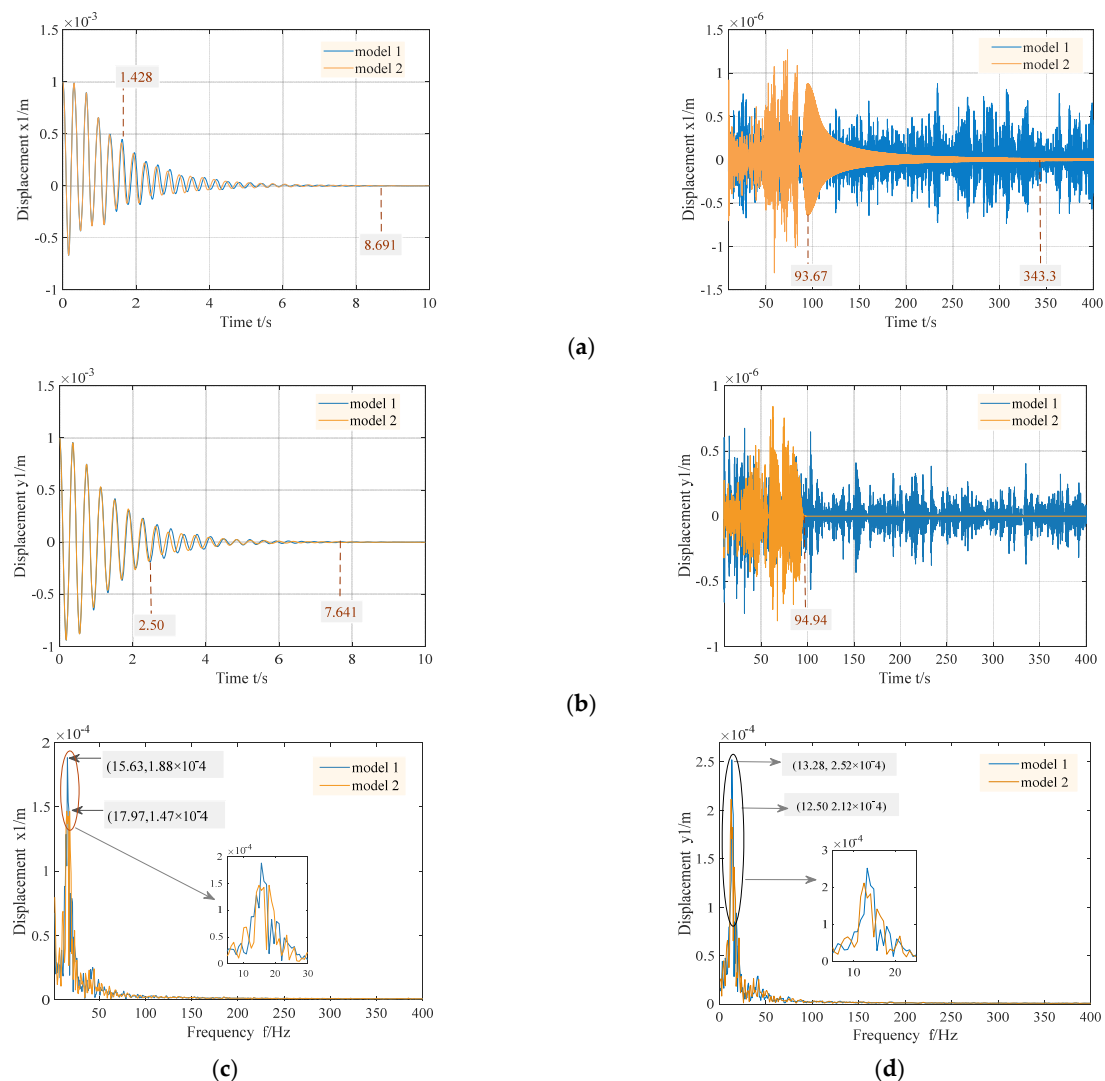


Figure 4. When considering hydraulic excitation, the time histories and frequency spectra of the generator rotor with or without the gyroscopic effect. (a) Temporal variation in the generator rotor's displacement along the x -direction (left graph shows 0~10 s, right graph shows 0~400 s); (b) temporal variation in the generator rotor's displacement along the y -direction (left graph shows 0~10 s, right graph shows 0~400 s); (c) frequency spectrum of the generator rotor's motion along the x -direction; (d) frequency spectrum of the generator rotor's motion along the y -direction.

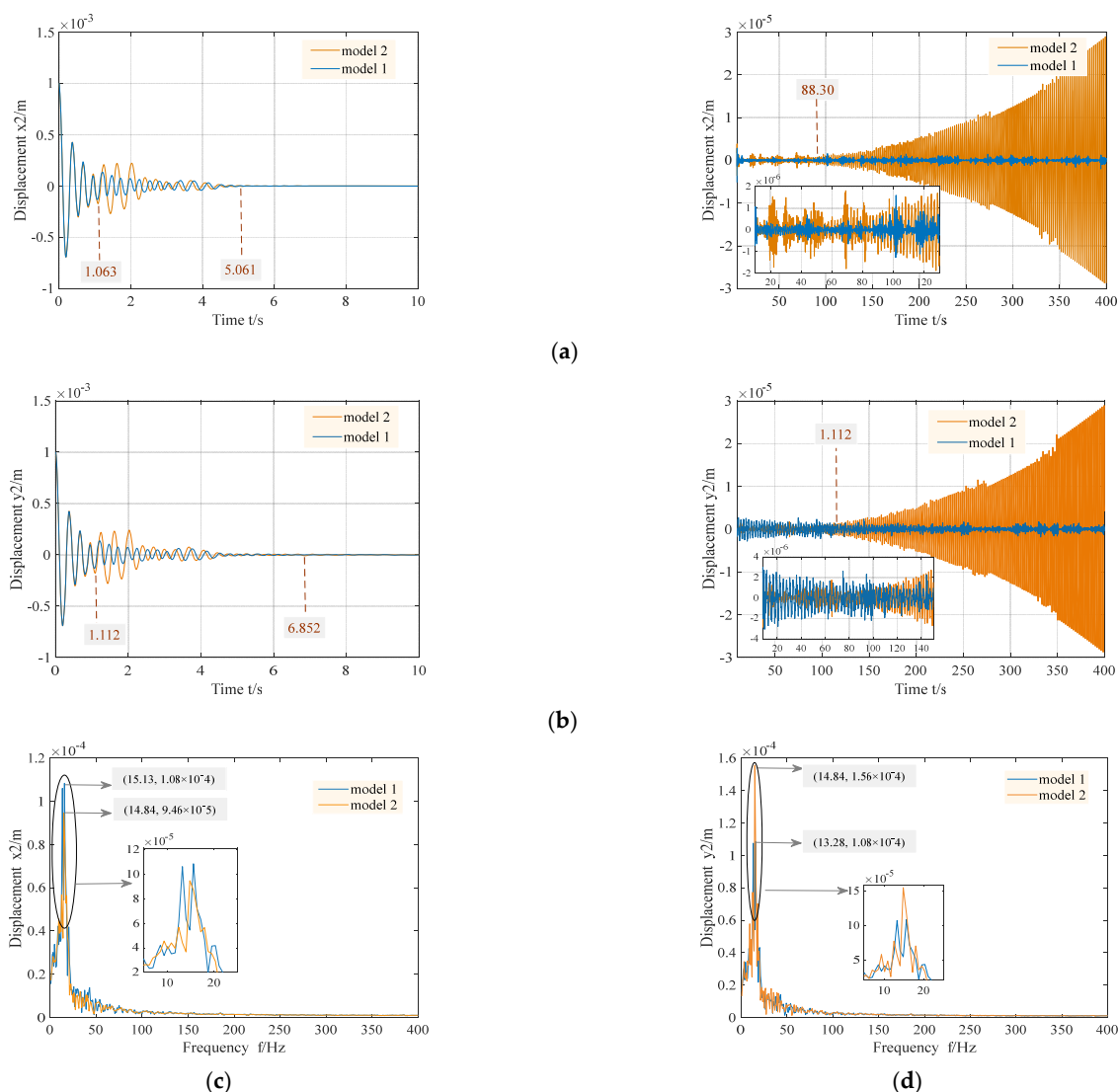


Figure 5. When considering hydraulic excitation, the time histories and frequency spectra of the hydro-turbine’s behavior with or without the gyroscopic effect. (a) Time history of the hydro–turbine’s motion along the x -axis (left graph shows 0~10 s, right graph shows 0~400 s); (b) time history of the hydro–turbine’s motion along the y -axis (left graph shows 0~10 s, right graph shows 0~400 s); (c) frequency spectrum of the hydro–turbine’s motion along the x -axis; (d) frequency spectrum of the hydro–turbine’s motion along the y -axis.

From Figure 4, the vibration time-series data for the generator rotor in both the x - and y -directions for model 1 and model 2 show minimal differences at the beginning of unit operation, and both demonstrate a trend of gradual decrease. The unit reached a relatively stable state after 7.641 s, the vibration amplitude remained in a relatively small range, and the vibration amplitude of the generator rotor in the x -direction gradually differed between 10 and 93.67 s. Following 93.67 s, the x -direction vibration amplitude of the generator rotor in model 2 (which does not account for the gyroscopic effect) underwent a gradual reduction. Conversely, the x -direction vibration amplitude of the generator rotor in model 1 (which incorporates the gyroscopic effect) did not exhibit a decreasing trend. Notably, after 110 s, the x -direction vibration amplitude of model 1 continued to surpass that of model 2. Similarly, in the y -direction vibration histories of the generator rotor, differences in vibration amplitude gradually emerged between 10 and 94.94 s, and the vibration amplitude rapidly decreased to a relatively small vibration range after 94.94 s, while in the subsequent operation, the vibration response of model 1 was always larger

than that of model 2. In addition, from an analysis of the vibration frequency spectrum, it is evident that under the influence of hydraulic excitation, the vibration amplitude of the generator rotor in the y -direction exceeds that in the x -direction. Initially, during the early stages of operation, the gyroscopic effect demonstrates a minimal impact on the vibration of the unit rotor. However, as the unit operates and the rotational speed increases to a specific threshold, the gyroscopic effect progressively intensifies the generator rotor's vibration, thereby compromising the operational stability of the HTGU.

From Figure 5, at the initial stage of unit operation, the vibration time histories of the turbine runner in both x - and y -directions for model 1 and model 2 exhibit negligible discrepancies, with the vibration amplitude stabilizing after 7.641 s. As illustrated in the second subplot of Figure 5, between 10 and 100 s, the x -direction vibration amplitudes of the turbine runners in both models remain within relatively stable ranges. Notably, the vibration amplitude of model 1 consistently remains lower than that of model 2 throughout this period. However, after 100 s, the vibration amplitude of model 2 gradually increases, but the vibration amplitude of model 1 still remains in a relatively small range. Similarly, it can be observed from the fourth subgraph of Figure 5 that the vibration amplitude of the turbine runner for model 1 in the y -direction is greater than that of model 2 between 10 and 100 s. After 100 s, the y -direction vibration amplitude of the turbine runner in model 2 gradually increases, while that of model 1 stays in a smaller vibration range with a slight decrease. Hence, it can be inferred that the impact of hydraulic excitation on the vibration characteristic of the turbine runner is not obvious in the early stage of operation. However, after the HTGU has operated for a certain duration, the impact of hydraulic excitation on the turbine runner becomes progressively more pronounced. In the absence of the gyroscopic effect, the vibration amplitude of the turbine runner demonstrates a gradual increase as the operation time elongates. This indicates that hydraulic excitation exerts an escalating influence on the turbine runner's dynamic behavior over extended operational periods when gyroscopic effects are not considered. But in the presence of the gyroscopic effect, the vibration amplitude of the turbine runner remains within a relatively narrow range. This phenomenon suggests that the gyroscopic effect mitigates the influence of hydraulic excitation on the turbine runner's vibration. Alternatively, it implies that hydraulic excitation has a negligible impact on the HTGU when the gyroscopic effect is considered.

3.2. Influence of Gyroscopic Effect on Bending–Torsional Coupling Vibration of the HTGU Without Hydraulic Excitation

Figures 6 and 7 represent the time histories and frequency spectra of the vibration of the generator rotor and turbine runner in the x - and y -directions without hydraulic excitation, (i.e., $F_{xyy} = 0$, $F_{yyy} = 0$, $M = 0$), respectively, in which model 1 indicates that the gyroscopic effect is considered and model 2 indicates that the gyroscopic effect is not considered.

From Figure 6, it can be seen that, in the early stages of operation, the generator rotor vibration amplitudes of model 1 and model 2 in both the x -direction and y -direction are notably high. Over time, however, these amplitudes decrease and stabilize within a significantly smaller range. The vibration amplitudes of the generator rotor in the x -direction of model 1 and model 2 are similar before 0–2.36 s. Following the initial stabilization phase ($t > 2.36$ s), comparative analysis reveals that model 2 exhibits marginally higher vibration amplitudes than model 1 in both the x - and y -directions. In addition, the vibration amplitudes of the generator rotors of model 1 and model 2 in the x -direction decrease rapidly after 17.08 s and stabilize in a smaller vibration range, and the vibration amplitudes of model 1 and model 2 in the x -direction are similar. Figure 6 illustrates that, within the first 2.02 s, model 1 and model 2 exhibit nearly identical y -direction vibration

amplitudes in their generator rotors. After the 2.02 s threshold, model 1 exhibits a subtle yet consistent increase in rotor vibration magnitude along the y -axis compared to model 2, with the difference becoming progressively clearer in later time segments. At 14.86 s, both model 1 and model 2 exhibit a sharp decline in rotor vibration magnitude along the y -axis, stabilizing thereafter within a significantly reduced amplitude range. A comparison of vibration spectra in the x - and y -directions indicates that model 1 generates stronger lateral displacements than model 2 during operation, particularly evident in the dominant frequency peaks of the x -axis response, when the hydraulic excitation is not considered. The spectral response analysis indicates that model 2 generates higher y -direction displacement magnitudes than model 1, with both models consistently displaying larger y -axis vibrations relative to their x -axis counterparts.

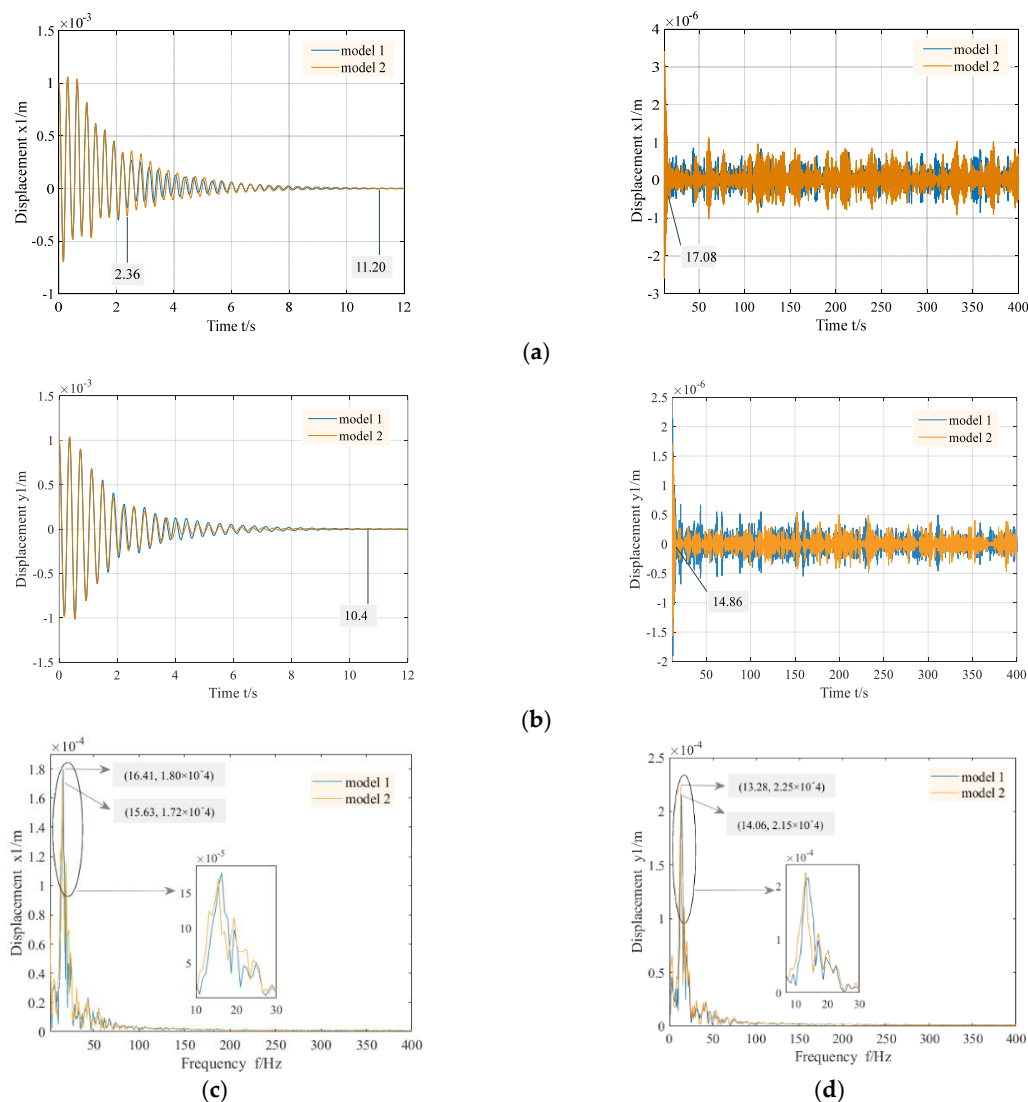


Figure 6. When hydraulic excitation is not considered, the time histories and frequency spectra of the generator rotor with or without the gyroscopic effect. (a) Time history of the hydro-turbine in the x -direction (left graph shows 0~10 s, right graph shows 0~400 s); (b) time history of the hydro-turbine in the y -direction (left graph shows 0~10 s, right graph shows 0~400 s); (c) frequency spectrum of the hydro-turbine in the x -direction; (d) frequency spectrum of the hydro-turbine in the y -direction.

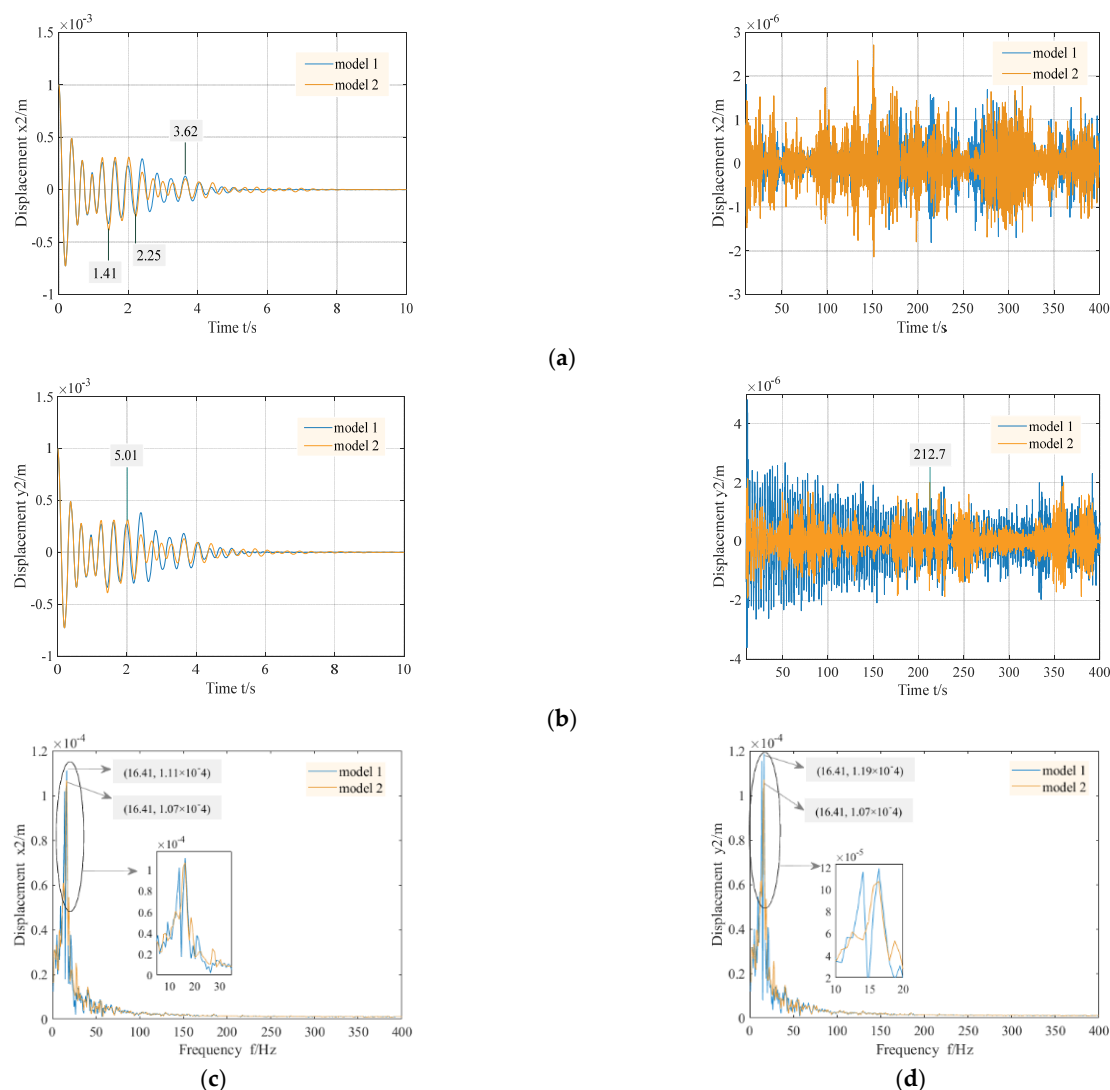


Figure 7. When hydraulic excitation is not considered, the time histories and frequency spectra of the hydro-turbine with or without the gyroscopic effect. **(a)** x -direction response of the hydro-turbine: **(left)** 0~10 s transient, **(right)** 0~400 s long-term behavior; **(b)** y -response of the hydro-turbine: **(left)** 0~10 s transient, **(right)** 0~400 s long-term behavior; **(c)** x -axis vibration frequency distribution of the hydro-turbine; **(d)** y -axis vibration frequency distribution of the hydro-turbine.

From the time histories of the vibration of the turbine runner in the x -direction in Figure 7, the spectral data confirm there are comparable x -direction vibration levels in the model 1 and model 2 turbine runners during the 0~1.41 s interval, excluding hydraulic excitation effects. Between 1.41 and 2.25 s, the comparative analysis reveals a shift in vibrational dominance between the models. While model 2 typically shows larger amplitudes, model 1 displays intensified vibrations during the 2.25~3.62 s period when examined through time-domain measurements. Beyond 10 s, model 1 and model 2 exhibit nearly identical x -axis vibrational displacements in the turbine runner, indicating convergence after initial divergence, and both remain in a relatively small vibration range. Figure 7's y -direction displacement profiles indicate comparable oscillation magnitudes for models 1 and 2 during the initial 5.01 s interval, demonstrating vibrational synchronization in the turbine runner. After 5.01 s, model 1 exhibits marginally greater oscillatory displacement in comparison to model 2, as evidenced by the peak amplitude measurements. Measurement data confirm that model 2 achieves a superior damping performance post-8.89 s, maintaining vertical vibrations below critical thresholds, while model 1 continues oscillating at higher magni-

tudes, with a tendency to decrease after 5.01 s until remaining in a relatively small vibration range after 212.7 s. In addition, spectral analysis of the turbine runner's vibrations reveals distinct modal characteristics on both orthogonal axes, with notable differences between horizontal (x) and vertical (y) dynamic responses. When there is no hydraulic excitation, model 2 maintains balanced vibrational characteristics across measurement directions, while model 1 exhibits significantly stronger dynamic responses in the vertical orientation than observed horizontally.

In order to describe the frequency spectra more intuitively, results are presented in table form in this paper. The vibrational displacements of both the generator rotor and turbine runner are presented in Table 2, with measurements taken along the orthogonal (x - and y -) axes under varying operational conditions.

Table 2. The vibration amplitude of the generator rotor and turbine runner in the x - and y -directions.

Directions	Scenarios	Vibration Amplitude (m)	
		Without Hydraulic Excitation	With Hydraulic Excitation
x_1	With gyroscopic effect	1.80×10^{-4}	1.88×10^{-4}
	Without gyroscopic effect	1.72×10^{-4}	1.47×10^{-4}
y_1	With gyroscopic effect	2.25×10^{-4}	2.52×10^{-4}
	Without gyroscopic effect	2.15×10^{-4}	2.12×10^{-4}
x_2	With gyroscopic effect	1.11×10^{-4}	1.08×10^{-4}
	Without gyroscopic effect	1.07×10^{-4}	9.46×10^{-5}
y_2	With gyroscopic effect	1.19×10^{-4}	1.08×10^{-4}
	Without gyroscopic effect	1.07×10^{-4}	1.56×10^{-4}

The comparative analysis in Table 3 reveals an observable increase in the torsional–flexural oscillations of the rotor assembly when hydraulic excitation is applied, particularly under the gyroscopic effect. However, when the gyroscopic effect is not considered, as shown in Table 3, the generator rotor experiences reduced bending–torsion coupled vibrations under hydraulic excitation compared to cases where such excitation is absent. Moreover, hydraulic excitation consistently reduces the coupled bending–torsion vibration amplitudes of the turbine runner in all cases—regardless of gyroscopic effects—with the sole exception being its y -directional motion. In contrast, the turbine runner experiences greater oscillatory displacement along the y -axis when subjected to hydraulic excitation than in unexcited conditions.

Table 3. Parameter design of flow rate, load current, and field current under different operating conditions.

Operating Condition	Flow Rate Q	Load Current i_j	Field Current F_{rm}, F_{sm}
1	960	1000	0.005
2	720	750	0.00375
3	480	500	0.0025
4	240	250	0.00125

3.3. Impact of Rotational Dynamics on Coupled Bending–Torsion Vibrations in Hydro-Turbine Generators Across Operational Regimes

To comprehensively analyze the dynamic behavior of hydro-turbine generators, this research employs multi-parameter simulations, varying flow rates, electrical loading, and

excitation currents across diverse operational scenarios. This comprehensive evaluation of dynamic behaviors under different working conditions provides a basis for identifying potential faults and abnormal vibration patterns. Table 3 presents the parameter configurations for four operating conditions, corresponding to full-load, high-load, medium-load, and low-load operation states.

This segment is organized using thematic subheadings to facilitate clarity. It presents a succinct yet rigorous analysis of the empirical findings, their analytical implications, and our key experimental inferences.

Figure 8 displays the vibration amplitude frequency-domain distribution of generator rotor and turbine runner displacements in the x - and y -directions when considering or neglecting the gyroscopic effect. The abscissa denotes the oscillation magnitude for both the generator's rotating element and the hydraulic runner along the orthogonal x - and y -axes, whereas the ordinate displays normalized occurrence rates. When examining how rotational inertia impacts generator oscillations, distinct patterns emerge in the magnitude distribution between orthogonal (x/y) axes. The gyroscopic forces produce markedly divergent vibration profiles across different spatial dimensions. As shown in Figure 8a, when neglecting the gyroscopic effect, the generator rotor vibration amplitude in both directions demonstrate highly concentrated characteristics, with approximately 90% of vibration energy clustered around the main frequency at approximately 1.2×10^{-5} m. In contrast, the model considering the gyroscopic effect (Figure 8b) exhibits typical normal distribution features. This transformation in distribution pattern indicates that the introduction of the gyroscopic effect causes separation between the forward and backward modes. The spectrum demonstrates complexity and randomness, which better align with the multi-factor coupling characteristics of actual generator rotor operation, resulting in a more uniform vibration amplitude distribution across different frequencies, whereas models neglecting the gyroscopic effect reflect more idealized conditions. Spectral analysis reveals that without considering the gyroscopic effect, the generator rotor vibration amplitude presents a single-peak characteristic, suggesting that the system operates in an approximately simple harmonic vibration state. When the gyroscopic effect is introduced, significant spectral broadening occurs, with the main frequency splitting into three characteristic peaks, caused by additional vibration modes induced by gyroscopic effects. Furthermore, in practice, there is anisotropic support stiffness due to structural or assembly factors in generator rotors. Gyroscopic moments can counteract or amplify these differences. When the gyroscopic effect is neglected, the negative half-axis frequency density in the y -direction exceeds that in the x -direction by 6.5% due to uncompensated local stiffness effects.

Analysis of the turbine runner vibration amplitude in the x - and y -directions shows significantly different characteristics. Figure 8c,d demonstrate that when the gyroscopic effect is neglected, turbine runner displacements—mainly determined by inherent structural characteristics and steady external loads—exhibit simpler dynamic behaviors, with approximately 90% of displacements in both directions concentrated around 0.01 mm, showing relatively stable vibration amplitude variations. With the gyroscopic effect considered, gyroscopic moments increase the system's equivalent stiffness, suppressing transverse vibrations caused by unbalance forces and other external excitation. This results in reduced displacement magnitudes for the turbine runner in both directions, shifted displacement peaks and main frequency ranges, and the formation of three characteristic peaks, reflecting increased vibration amplitude complexity.

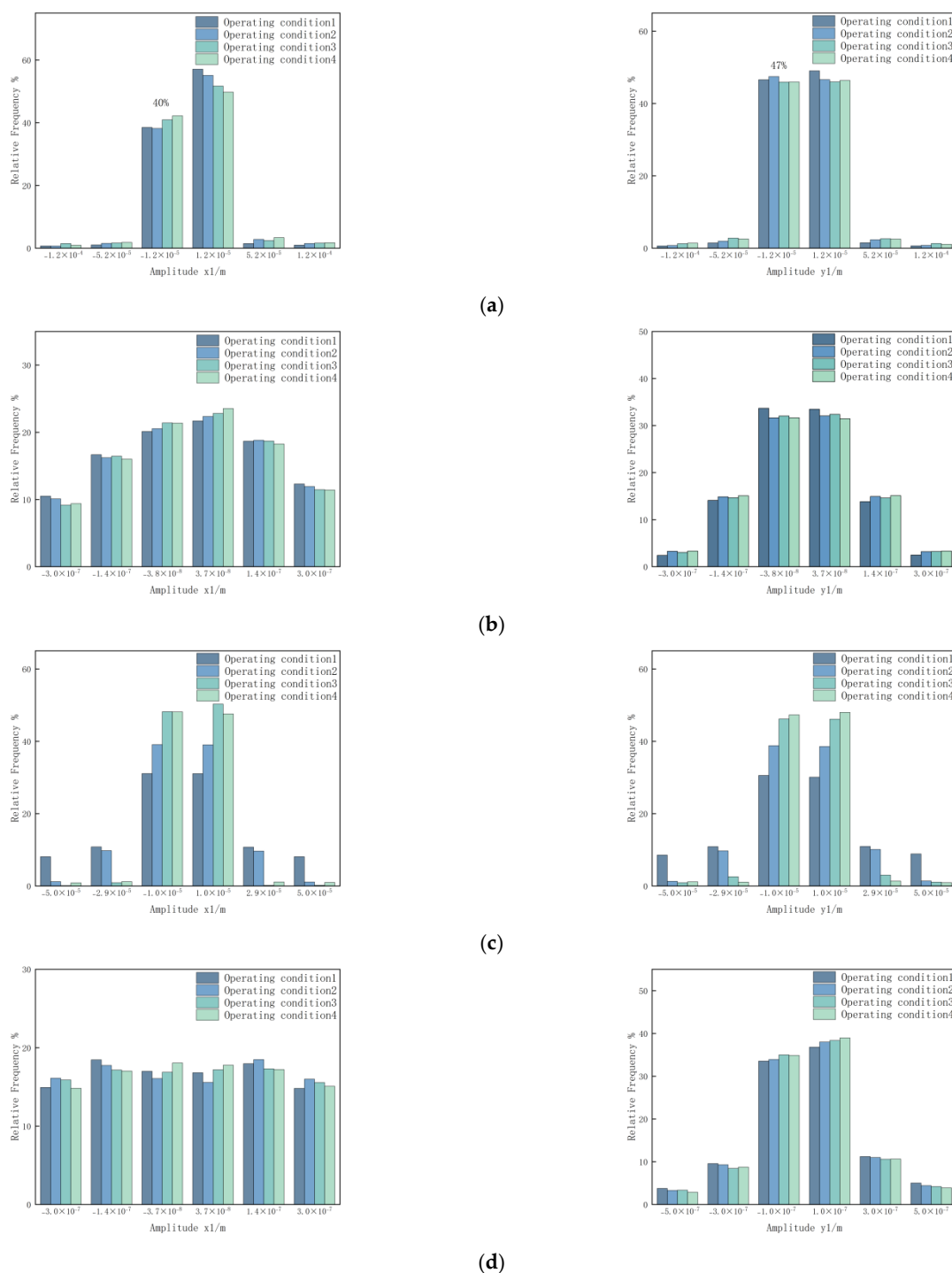


Figure 8. Vibration amplitude frequency-domain distribution of the generator rotor and hydro-turbine with or without the gyroscopic effect. Amplitude distribution of the generator rotor: (a) no gyroscopic effect, x -axis (left) and y -axis (right), (b) with gyroscopic effect, x -axis (left) and y -axis (right); amplitude distribution of the hydro-turbine: (c) no gyroscopic effect, x -axis (left) and y -axis (right), (d) with gyroscopic effect, x -axis (left) and y -axis (right).

Panels (a) and (b) of Figure 9 demonstrate how four distinct operational states affect the lateral displacement characteristics of both the generator's rotating element and hydraulic runner along perpendicular axes, excluding rotational inertia effects. Both components show significant initial displacement fluctuations that gradually stabilize, explained by large initial imbalanced forces during turbine startup, which dissipate through damping effects over time. Additionally, axial fluid flow generates vortex-induced vibration and

pressure pulsation, which directly excite greater displacements in the y -direction compared to the x -direction. A comparison of Figure 9b,d reveals that generator rotors—typically featuring higher rotational speeds and more complex structures—experience generally larger displacement fluctuations than turbine runners due to their greater inherent imbalanced masses.

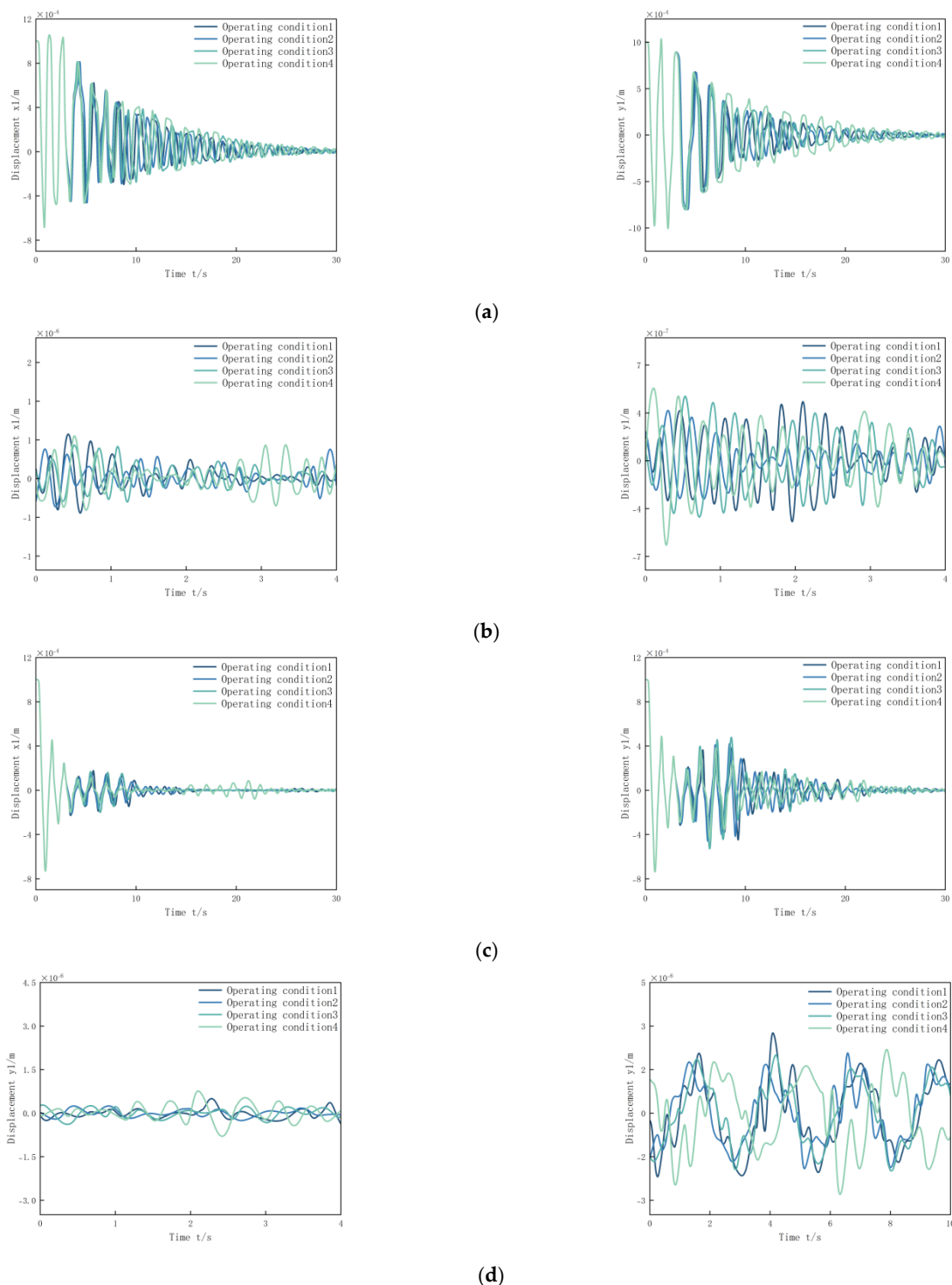


Figure 9. Time histories of the generator rotor and hydro-turbine with or without the gyroscopic effect. Generator rotor (a) without (left figure x -direction, right figure y -direction) and (b) with the gyroscope effect (left figure x -direction, right figure y -direction); hydro-turbine (c) without (left figure x -direction, right figure y -direction) and (d) with the gyroscope effect (left figure x -direction, right figure y -direction).

Figure 9c,d present the influence of four operating conditions on displacements when considering the gyroscopic effect. All conditions exhibit periodic fluctuations in the 10^{-6} – 10^{-7} m range, with Condition 4 showing relatively larger fluctuations, indicating the highest instability. Condition 2 and Condition 3 demonstrate consistent fluctuation frequencies and vibration amplitudes, suggesting equivalent gyroscopic effects. Condition 1 displays relatively smaller fluctuations and smoother curves, indicating a minimal gyroscopic influence on the turbine runner and higher system stability under this condition. However, for the generator rotor, smaller y -direction displacements compared to the x -direction suggest a weaker gyroscopic influence in the vertical direction. Temporally, all conditions show stable periodic fluctuations within 0–4 s without noticeable decay or growth, indicating a dynamic equilibrium. Synchronized peak occurrences across conditions reveal synchronized dynamic responses of the generator rotor in both directions under various operating conditions.

4. Conclusions and Discussion

This research investigates how hydraulic disturbances and rotational inertia jointly affect combined bending and twisting oscillations in hydroelectric units. A specialized non-linear mathematical framework incorporating both fluid-induced excitation and Coriolis coupling effects was developed for this coupled vibration analysis. The following conclusions and suggestions are drawn from comparative studies. Firstly, the gyroscopic effect has no significant impact on the bending–torsion vibration of the hydroelectric turbine generator set during the initial stage (0–2 s) of operation. The difference in the vibration amplitude of the generator rotor in the x -direction between the two models is only 3.2%. However, after the unit has been running for a period of time, as the rotational speed increases, the effect of the gyroscopic effect on the bending–torsional vibration of the unit gradually becomes apparent. The amplitude difference in the x -direction of the generator rotor is 28%, and in the y -direction is 19%. In model 2, the amplitude in the y -direction of the turbine runner is 44% higher than that in model 1, while the amplitude in the x -direction is 12% lower than that in model 1. Therefore, it affects the operational stability of the HTGU. Secondly, when considering the gyroscopic effect, the amplitude of the bending–torsion coupling vibration of the generator rotor with hydraulic excitation is greater than that of the generator rotor without hydraulic excitation. The generator rotor with hydraulic excitation is elevated by 4.4% in the x -direction and 12% in the y -direction compared to the case without hydraulic excitation. However, when the gyroscopic effect is not taken into account, the bending–torsion coupling vibration amplitude of the generator rotor with hydraulic excitation is smaller than that of the generator rotor without hydraulic excitation. The x -direction vibration amplitude of the generator rotor with hydraulic excitation is reduced by 14.5% compared to the case without hydraulic excitation, and the y -direction amplitude decreases by only 1.4%. Thirdly, for the generator rotor, the presence or absence of hydraulic excitation has no influence on the vibration of model 1 (with the gyroscopic effect), but the vibration of model 2 (without the gyroscopic effect) is greatly affected. However, due to the limitations of the experimental conditions, this paper only presents the theoretical calculation results. In the future, we will further verify the reliability of this model through real experimental data.

Author Contributions: Conceptualization, Z.B. and J.L.; data curation, Y.M., X.S., H.S. and P.Z.; formal analysis, X.S. and H.S.; investigation, Z.B.; methodology, Z.B., J.L., Y.M. and X.S.; project administration, H.S., P.Z., B.P. and N.X.; resources, P.Z., X.L., S.G., B.P., N.X. and C.S.; software, J.L., Y.M., X.S., X.L. and S.G.; supervision, Z.Z., C.S. and D.C.; visualization, Y.Y.; writing—original draft, Z.B. and J.L.; writing—review & editing, Z.Z. and D.C. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the Open Research Fund of the Hubei Technology Innovation Center for Smart Hydropower (Grant No. 1523020038).

Data Availability Statement: The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

List of symbols

c_1	Damping coefficient of generator rotor
c_2	Damping coefficient of turbine runner
$c_{r\theta}$	Structural damping coefficient of inclined vibration
c_t	Structural damping coefficient of torsional vibration
J_d	Diameter moment of inertia
J_{p1}	Polar moment of inertia of generator rotor
J_{p2}	Polar moment of inertia of turbine runner
R	Radius of generator rotor
r_1	Distance from runner inlet to runner axis
r_2	Distance from runner outlet to runner axis
L	Length of generator rotor

References

1. Sun, W.; Wang, B.; Ma, Z.; Zhang, H. Insights into the nonlinear dynamic characteristics of a water turbine generator set considering the coupled effects of a fractional-order broadband foundation. *J. Sound Vib.* **2024**, *573*, 118190. [\[CrossRef\]](#)
2. Chen, W.; Wang, S.; Chen, H.; Zhang, W.; Ma, J.; Pan, J. Analysis of Shafting System Vibration Characteristics for Mixed-Flow hydropower Units Considering Sand Wear on Turbine Blades. *J. Appl. Sci.* **2024**, *14*, 4806. [\[CrossRef\]](#)
3. Zhang, J.; Ma, Z.; Wang, X.; Wu, Q.; Zhang, L. Transient vibration of shafting in coupled hydraulic-mechanical-electrical-structural system for hydropower station during start-up process. *Appl. Math. Model.* **2023**, *124*, 860–880. [\[CrossRef\]](#)
4. Xu, Y.; Li, Z.H. Computational Model for Investigating the Influence of Unbalanced Magnetic Pull on the Radial Vibration of Large Hydro-Turbine Generators. *J. Vib. Acoust.* **2012**, *134*, 051013. [\[CrossRef\]](#)
5. Wang, B.; Xue, J.Y.; Wu, F.J.; Zhu, D.L. Finite time takagi-sugeno fuzzy control for hydro-turbine governing system. *J. Vib. Control* **2018**, *24*, 1001–1010. [\[CrossRef\]](#)
6. Wang, Z.; Zhu, C.C. A new model for analyzing the vibration behaviors of rotor-bearing system. *Commun. Nonlinear Sci. Numer. Simul.* **2020**, *83*, 105130. [\[CrossRef\]](#)
7. Nie, L.; Wang, X.; Zhang, J.; Ma, Z.; Zhang, L. Dynamic characteristics analysis and vibration control of coupled bending-torsional system for axial flow hydraulic generating set. *Chaos Solitons Fractals* **2024**, *187*, 115401. [\[CrossRef\]](#)
8. Shi, Y.; Shen, J.; Guo, W.; Li, C.; Zheng, Y.; Zhao, Z.; Zhou, J.; Zhang, Y. The nonlinear dynamic characteristics analysis and vibration reduction control of the shafting system of a hydropower unit. *Int. J. Non-Linear Mech.* **2022**, *146*, 104166. [\[CrossRef\]](#)
9. Zhang, J.; Ma, Z.; Wang, X.; Zhang, L.; Wu, Q.; Li, M. Vibration control on coupled unit-plant structure of pumped storage power station during sudden load-up process. *Mech. Syst. Signal Process.* **2024**, *212*, 111333. [\[CrossRef\]](#)
10. Zhu, D.; Tao, R.; Xiao, R.F.; Pan, L.T. Solving the Runner Blade Crack Problem for a Francis Hydro-Turbine Operating under Condition-Complexity. *Renew. Energy* **2020**, *149*, 298–320. [\[CrossRef\]](#)
11. Song, Z.Q.; Liu, Y.H.; Guo, P.C.; Feng, J.J. Torsional Vibration Analysis of Hydro-Generator Set Considered Electromagnetic and Hydraulic Vibration Resources Coupling. *Int. J. Precis. Eng. Manuf.* **2018**, *19*, 939–945. [\[CrossRef\]](#)
12. Perez-Loya, J.J.; Abrahamsson, C.J.D.; Lundin, U. Electromagnetic Losses in Synchronous Machines During Active Compensation of Unbalanced Magnetic Pull. *IEEE Trans. Ind. Electron.* **2018**, *66*, 124–131. [\[CrossRef\]](#)
13. Wu, X.Y.; Peng, Z.K.; Ren, J.S.; Cheng, C.M.; Zhanga, W.; Wang, D. Rub-Impact Fault Diagnosis of Rotating Machinery Based on 1-D Convolutional Neural Networks. *IEEE Sens. J.* **2020**, *20*, 8349–8363. [\[CrossRef\]](#)
14. Yang, Y.; Dong, X.J.; Peng, Z.K.; Zhang, W.M.; Meng, G. Vibration signal analysis using parameterized time-frequency method for features extraction of varying-speed rotary machinery. *J. Sound Vib.* **2015**, *335*, 350–366. [\[CrossRef\]](#)
15. Xu, B.B.; Chen, D.Y.; Zhang, H.; Li, C.; Zhou, J.Z. Shaft mis-alignment induced vibration of a hydraulic turbine generating system considering parametric uncertainties. *J. Sound Vib.* **2018**, *435*, 74–90. [\[CrossRef\]](#)

16. Xu, Y.; Li, Z.H.; Lai, X.D. Dynamic model for hydro-turbine generator units based on a database method for guide bearings. *Shock Vib.* **2013**, *20*, 411–421. [[CrossRef](#)]
17. Zhou, W.; Ma, J.; Ma, Z.; Yu, W.; Su, H.; Gao, B. Fluid–structure interaction on the rotor-dynamic characteristics of a low-specific-speed centrifugal pump considering multi-scale fluid excitation effects. *Phys. Fluids* **2024**, *36*, 117157. [[CrossRef](#)]
18. Shi, Y.S.; Zhou, J.Z.; Lai, X.J.; Xu, Y.H.; Guo, W.C.; Liu, B.N. Stability and sensitivity analysis of the bending-torsional coupled vibration with the arcuate whirl of hydro-turbine generator unit. *Mech. Syst. Signal Process.* **2020**, *149*, 107306. [[CrossRef](#)]
19. Sun, W.; Guo, Z. Mathematical modeling and nonlinear vibration analysis of a coupled hydro-generator shaft-foundation system. *Commun. Nonlinear Sci. Numer. Simul.* **2021**, *98*, 105776. [[CrossRef](#)]
20. Hua, C.L.; Cao, G.H.; Rao, Z.S.; Ta, N.; Zhu, Z.C. Coupled bending and torsional vibration of a rotor system with nonlinear friction. *J. Mech. Sci. Technol.* **2017**, *31*, 2679–2689. [[CrossRef](#)]
21. Zhang, W.Y.; Zhu, P.F.; Wang, J.P.; Zhu, H.Q. Stability Control for a Centripetal Force Type-Magnetic Bearing-Rotor System Based on Golden Frequency Section Point. *IEEE Trans. Ind. Electron.* **2021**, *68*, 12482–12492. [[CrossRef](#)]
22. Han, B.C.; Chen, Y.L.; Li, M.X.; Zheng, S.Q.; Zhang, X. Stable Control of Nutation and Precession for the Radial Four-Degree-of-Freedom AMB-Rotor System Considering Strong Gyroscopic Effects. *IEEE Trans. Ind. Electron.* **2021**, *68*, 11369–11378. [[CrossRef](#)]
23. Zhao, J.S.; Liu, W.T.; Zhang, Y.; Feng, Z.J. Effects of gyroscopic moment on the damage of a tapered roller bearing. *Mech. Mach. Theory* **2013**, *69*, 185–199. [[CrossRef](#)]
24. Jalili, M.M.; Hesabi, J.; Abootorabi, M.M. Simulation of forced vibration in milling process considering gyroscopic moment and rotary inertia. *Int. J. Adv. Manuf. Technol.* **2016**, *89*, 2821–2836. [[CrossRef](#)]
25. She, H.X.; Li, C.F.; Tang, Q.S.; Wen, B.C. Influence mechanism of disk position and flexibility on natural frequencies and critical speeds of a shaft-disk-blade unit. *J. Sound Vib.* **2020**, *469*, 115156. [[CrossRef](#)]
26. Zeng, Y.; Zhang, L.X.; Zhang, C.L.; Qian, J.; Guo, Y.K. Large disturbance transient model for shafting lateral vibration of Hydro Turbine Generation sets. *China J. Solid Mech.* **2013**, *33*, 137–142. (In Chinese)
27. Zeng, Y.; Zhang, L.X.; Guo, Y.K.; Qian, J.; Zhang, C.L. The generalized Hamiltonian model for the shafting transient analysis of the hydro turbine generating sets. *Nonlinear Dyn.* **2014**, *76*, 1921–1933. [[CrossRef](#)]
28. Xu, B.B.; Chen, D.Y.; Tolo, S.; Patelli, E.; Jiang, Y.L. Model validation and stochastic stability of a hydro-turbine governing system under hydraulic excitations. *Int. J. Electr. Power Energy Syst.* **2018**, *95*, 156–165. [[CrossRef](#)]
29. Zhang, L.; Ma, Z.; Song, B. Dynamic characteristics of a rub-impact rotor-bearing system for hydraulic generating set under unbalanced magnetic pull. *Arch. Appl. Mech.* **2013**, *83*, 817–830. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.